

Badr ALKHAMISSI

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https://bkhmsi.github.io github.com/bkhmsi linkedin.com/in/bkhmsi scholar.google.com/bkhmsi

I am passionate about building intelligent systems that can solve complex machine learning tasks and at the same time reveal insights about the inner workings of biological agents. Such systems do not only pave a way for solving long-standing societal problems but also shed light on what it means to be human.

Please see [website](#) for a more detailed resumé.

EDUCATION

Present September 2023	EPFL, PHD, Lausanne, Switzerland ➤ Co-advised by Antoine Bosselut (NLP Lab) and Martin Schrimpf (NeuroAI Lab).
September 2020 September 2019	Goldsmiths, University of London, MASTER OF SCIENCE, London, UK ➤ MSc. in Computational Cognitive Neuroscience - Pass with Distinction. ➤ Thesis: Episodic Meta-Reinforcement Learning on the Two-Step and Harlow Tasks (Grade: 90%).
June 2018 September 2014	The American University in Cairo, BACHELOR OF SCIENCE, Cairo, Egypt ➤ BSc. in Computer Science - Summa Cum Laude. GPA 4.0 ➤ Thesis: Video Search via Natural Language.

SELECTED RESEARCH AND INDUSTRIAL EXPERIENCE

Present July 2026	Student Researcher, GOOGLE DEEPMIND, Zurich, Switzerland ➤ Currently a student researcher at Google DeepMind. Machine Learning Latent Reasoning Looped Transformers AI for Science
November 2022 November 2021	AI Resident, META AI, Seattle, US ➤ Research in NLP. Published 5 first-author papers, and 1 co-author paper. (See Publications) NLP Hate Speech Responsible AI Reasoning Arabic NLP FAIR
October 2021 October 2020	Research Intern, SONY AI, Tokyo, Japan (Remotely) ➤ Published 2 papers in collaboration with DeepMind and 1 paper with Sony CSL. (See Publications) Meta-learning Reinforcement Learning Symbolic Alchemy
September 2019 April 2019	Applied Scientist, MICROSOFT ADVANCED TECHNOLOGY LAB IN CAIRO, Cairo, Egypt ➤ Part of the LUIS Applied Science Team and the Machine Teaching Innovation Group. NLU NLP MT Keras Tensorflow PyTorch Python C#

SELECTED PUBLICATIONS

Selected Conferences

APTURUS: DEMOCRATIZING OPEN AND COMPLIANT LLMs FOR GLOBAL LANGUAGE ENVIRONMENTS  Paper  Media  Models  Tweets Authors: Alejandro Hernández-Cano, Alexander Hägele, [...], Badr AlKhamissi , [...], Antoine Bosselut*, Imanol Schlag*, Martin Jaggi*	ACL 2026
MIXTURE OF COGNITIVE REASONERS: MODULAR REASONING WITH BRAIN-LIKE SPECIALIZATION  Paper  Website  Tweets  github.com/https://github.com/bkhmsi/mixture-of-cognitive-reasoners Authors: Badr AlKhamissi , C. Nicolò De Sabbata, Zeming Chen, Martin Schrimpf*, Antoine Bosselut*	ICLR 2026
INDUCING DYSLEXIA IN VISION LANGUAGE MODELS  Paper  Tweets  github.com/https://github.com/epflneuroailab/VWFA-Localization Authors: Melika Honarmand, Ayati Sharma, Badr AlKhamissi , Johannes Mehrer*, Martin Schrimpf*	ICLR 2026
HIRE YOUR ANTHROPOLOGIST! RETHINKING CULTURE BENCHMARKS THROUGH AN ANTHROPOLOGICAL LENS  Paper  Tweets Authors: Mai AlKhamissi*, Yunze Xiao*, Badr AlKhamissi , Mona Diab	EACL 2026
FROM LANGUAGE TO COGNITION: HOW LLMs OUTGROW THE HUMAN LANGUAGE NETWORK Oral  Paper  Tweets Authors: Badr AlKhamissi , Greta Tuckute, Yingtian Tang, Taha Binhuraib, Antoine Bosselut*, Martin Schrimpf*	EMNLP 2025

THE LLM LANGUAGE NETWORK: A NEUROSCIENTIFIC APPROACH FOR IDENTIFYING CAUSALLY IMPORTANT UNITS	NAACL 2025
Oral Paper Website github.com/https://github.com/bkhmsi/llm-localization Authors: Badr AlKhamissi , Greta Tuckute, Antoine Bosselut*, Martin Schrimpf*	
TOPOLM: BRAIN-LIKE SPATIO-FUNCTIONAL ORGANIZATION IN A TOPOGRAPHIC LANGUAGE MODEL	ICLR 2025
Oral Paper Tweets Authors: Neil Rathi*, Johannes Mehrer*, Badr AlKhamissi , Taha Binhuraib, Nicholas M. Blauch, Martin Schrimpf	
INVESTIGATING CULTURAL ALIGNMENT OF LARGE LANGUAGE MODELS	ACL 2024
Paper Tweets github.com/https://github.com/bkhmsi/cultural-trends Authors: Badr AlKhamissi , Muhammad ElNokrashy, Mai AlKhamissi, Mona Diab	
A CONTEXT-CONTRASTIVE INFERENCE APPROACH TO PARTIAL DIACRITIZATION	ARABICNLP 2024
Oral Paper Demo github.com/https://github.com/munael/ccpd-partial-arabic-diacritization Authors: Muhammad ElNokrashy, Badr AlKhamissi	
INSTRUCTION-TUNING ALIGNS LLMs TO THE HUMAN BRAIN	COLM 2024
Paper Tweets Authors: Khai Loong Aw, Syrielle Montariol*, Badr AlKhamissi *, Martin Schrimpf, Antoine Bosselut	
DEPTH-WISE ATTENTION (DWATT): A LAYER FUSION METHOD FOR DATA-EFFICIENT CLASSIFICATION	LREC-COLING 2024
Oral Paper Poster Authors: Muhammad ElNokrashy*, Badr AlKhamissi *, Mona Diab	
“FLEX TAPE CAN’T FIX THAT”: BIAS AND MISINFORMATION IN EDITED LANGUAGE MODELS	EMNLP 2024
Paper Authors: Karina Halevy, Anna Sotnikova, Badr AlKhamissi , Syrielle Montariol, Antoine Bosselut	
ALERT: ADAPT LANGUAGE MODELS TO REASONING TASKS	ACL 2023
Paper Poster Authors: Ping Yu, Tianlu Wang, Olga Golovneva, Badr AlKhamissi , Siddharth Verma, Zhijing Jin, Gargi Ghosh, Mona Diab, Asli Celikyilmaz	
TOKEN: TASK DECOMPOSITION AND KNOWLEDGE INFUSION FOR FEW-SHOT HATE SPEECH DETECTION	EMNLP 2022
Oral Paper Tweets Authors: Badr AlKhamissi *, Faisal Ladhak*, Srinu Iyer, Ves Stoyanov, Zornitsa Kozareva, Xian Li, Pascale Fung, Lambert Mathias, Asli Celikyilmaz, Mona Diab	

Selected Workshops

A SELF-SYNTHESIS APPROACH FOR TRAINING VISION-LANGUAGE MODELS WITH DEVELOPMENTALLY PLAUSIBLE DATA	BABYLM 2024
CoNLL 2024 Paper Authors: Badr AlKhamissi *, Yingtian Tang*, Abdülkadir Gökçe*, Johannes Mehrer, Martin Schrimpf	
KHATTAT: ENHANCING READABILITY AND CONCEPT REPRESENTATION OF SEMANTIC TYPOGRAPHY	AI4VA 2024
ECCV 2024 Paper Website Tweets Authors: Ahmed Hussein*, Alaa Elsetohy*, Sama Hadhoud*, Tameem Bakr*, Yasser Rohaim*, Badr AlKhamissi	
OPT-R: EXPLORING THE ROLE OF EXPLANATIONS IN FINETUNING AND PROMPTING FOR REASONING SKILLS OF LLMs	NLRSE 2023
ACL 2023 Paper Poster Tweets Authors: Badr AlKhamissi , Siddharth Verma, Ping Yu, Zhijing Jin, Asli Celikyilmaz, Mona Diab	
MULTITASK LEARNING WITH SELF-CORRECTION FOR HATE SPEECH CLASSIFICATION	OSACT5 2022
LREC 2022 Oral Paper Best Paper Award Tweets Authors: Badr AlKhamissi *, Mona Diab	
HOW TO LEARN AND REPRESENT ABSTRACTIONS: AN INVESTIGATION USING SYMBOLIC ALCHEMY	AUTOML WORKSHOP 2022
AutoML 2022 Paper Poster github.com/https://github.com/BKHMSI/alchemist Authors: Badr AlKhamissi , Akshay Srinivasan, Zeb Kurth-Nelson, Sam Ritter	
THE EMERGENCE OF ABSTRACT AND EPISODIC NEURONS IN EPISODIC META-RL	MEMARI 2022
NeurIPS 2022 Spotlight Oral Paper Poster Authors: Badr AlKhamissi , Muhammad ElNokrashy, Michael Spranger	

Selected Preprints

DISCOVERING FUNCTIONALLY SELECTIVE BRAIN REGIONS WITH A DEEP TOPOGRAPHIC MULTIMODAL MODEL	PREPRINT 2026
Paper Website github.com/https://github.com/epflneuroailab/topo-omni Tweets Authors: Badr AlKhamissi *, Johannes Mehrer*, Lara Marinov, Ahmed Abdelaal, Abdulkadir Gokce, Martin Schrimpf	
MEDITRON: OPEN MEDICAL FOUNDATION MODELS ADAPTED FOR CLINICAL PRACTICE	PREPRINT 2024
Paper github.com/https://github.com/epfLLM/meditron Authors: Zeming Chen, Angelika Romanou*, Antoine Bonnet*, Alejandro Hernández-Cano*, Badr AlKhamissi *, Kyle Matoba*, [...], Physician Evaluation Group, Noémie Boillat-Blanco, Kristina Keitel, Javier Elkin, Blaise Robert, Syrielle Montariol, Mary-Anne Hartley*, Martin Jaggi*, Antoine Bosselut*	

SELECTED AWARDS

- > KAUST Rising Stars in AI Recipient. '26
- > Swiss AI Initiative PhD Fellowship Recipient. '25
- > EPFL Doc.Mobility Grant Recipient. '25
- > President's Cup and Mohamed El Beledy Academic Award for class valedictorian - American University in Cairo '18.
- > Mohammed Bin Abdulkarim A. Allehedan Award for highest performing STEM student - American University in Cairo '18.
- > First place on the two categories of the Arabic Reverse Dictionary shared task at ArabicNLP 2023. '23
- > Best Paper Award for the Fine-Grained Hate Speech Detection shared task in Arabic. '22
- > First place in all four categories of the second NADI shared task. '21
- > Best Mentor Award at Microsoft Advanced Technology Lab in Cairo. '19
- > Shortlisted in Dell EMC's Graduation Project Competition - American University in Cairo '18
- > Computer Science Exemplary Student - American University in Cairo '18
- > Outstanding Performance Award in Deep Learning Summer School at Tsinghua University. '17

TEACHING EXPERIENCE

Spring '24,'25,'26 & Fall '24	Teaching Assistant, EPFL, Lausanne, Switzerland > Teaching assistant for Modern Natural Language Processing and Linear Algebra. Python PyTorch NLP LLMs RLHF
Fall '15 & '17 & Spring '16	Undergraduate Teaching Assistant, THE AMERICAN UNIVERSITY IN CAIRO, Cairo, Egypt > Teaching assistant for Deep Learning and Programming Fundamentals courses. C++ Python Tensorflow Keras SFML

SELECTED INVITED TALKS

- > IndabaX Sudan Featured Speaker: Building Responsible & Ethical LLMs.
- > LauzHack Deep Learning Bootcamp: Brain-Inspired Large Language Models.
- > REST-CL Retreat for Students in Computational Linguistics (UPF University) The LLM Language Network.
- > Psychology Days at Riga Stradins University: Vision and Language in Brains, Minds and Machines.
- > Nancy Kanwisher's Lab at MIT: Mixture of Cognitive Reasoners.
- > NCCR Evolving Language Group: Brain Alignment of Untrained Models.
- > Cohere For AI Community Talk (Geo Regional Africa Group): Investigating Cultural Alignment of LLMs.
- > ARBML Board Talk: Investigating Cultural Alignment of LLMs.
- > AiBelmasry Podcast (Guest Speaker): Analyzing LLM's Ability in Reasoning.
- > TEDxMIU Talk: Representation Matters: Celebrating Egyptians in the AI Revolution.
- > Cohere For AI Community Talk: Presenting OPT-R Paper.
- > Mada Masr Podcast Guest Speaker (April 6th 2023 Episode): A New World Shaped by Artificial Stupidity.
- > IndabaX Egypt 2019 Jointly led a tutorial on cross-lingual alignment.

SERVICE

ORGANIZING COMMITTEE

ICLR 2026 Workshop on Representational Alignment (Re-Align). ArabicNLP 2025 (Publicity Chair). ICML Workshop on Large Languages and Cognition (2024). ArabicNLP 2024 (Social Chair).

JOURNAL REVIEWER & SELECTED PROGRAM COMMITTEE MEMBER (REVIEWER)

Nature Communications. Neurobiology of Language. Nature Machine Intelligence. TACL. OpenMind. EMNLP 2026 (Area Chair). CogSci 2026. EACL 2025. NeurIPS 2025. EMNLP 2025. NAACL 2025 (Area Chair). EMNLP 2024. NeurIPS 2024. ArabicNLP 2024. CCN 2024. ACL 2024. NAACL 2024. EMNLP 2023. ACL 2023. ENLSP-II Workshop at NeurIPS 2022. EMNLP 2022. AutoML 2022. ICBINB at NeurIPS 2021. Meta-Learn at NeurIPS 2021, 2020. L2L at ICLR 2021.